The heat operator on the big real interpolation space does not have global L^∞-maximal regularity

Solution packet for arXiv:2502.16521, Remark 3.4(c)

11 August 2026

Candidate full negative answer. For every $n \geq 1$ and every $0 < \theta < 1$, the part of $-\Delta$ in $(C_0(\mathbb{R}^n), \mathrm{D}(-\Delta))_{\theta,\infty}$ does *not* have L^∞-maximal regularity on $\mathbb{R}_+$.

1 The question

The source is Sebastian Król, Mieczysław Mastyło, and Jarosław Sarnowski, *Maximal regularity estimates for the abstract Cauchy problems*, arXiv:2502.16521v1 [1]. On PDF page 40, Remark 3.4(c) asks:

> *We do not know whether for the negative Laplacian $\mathcal{A} := -\Delta$ on $\mathcal{X} = C_0(\mathbb{R}^n)$ its part in $X := (\mathcal{X}, D_{\mathcal{A}})_{\theta,\infty}$, $\theta \in (0,1)$, has L^∞-m.r. on the whole $\mathbb{R}_+$.*

Here $\mathcal{A} = -\Delta$ has domain

$$\mathrm{D}(\mathcal{A}) = \{f \in C_0(\mathbb{R}^n) : -\Delta f \in C_0(\mathbb{R}^n) \text{ in the distributional sense}\},$$

and $H_t = e^{-t\mathcal{A}} = e^{t\Delta}$ denotes the heat semigroup. If

$$X_\theta = (C_0(\mathbb{R}^n), \mathrm{D}(\mathcal{A}))_{\theta,\infty},$$

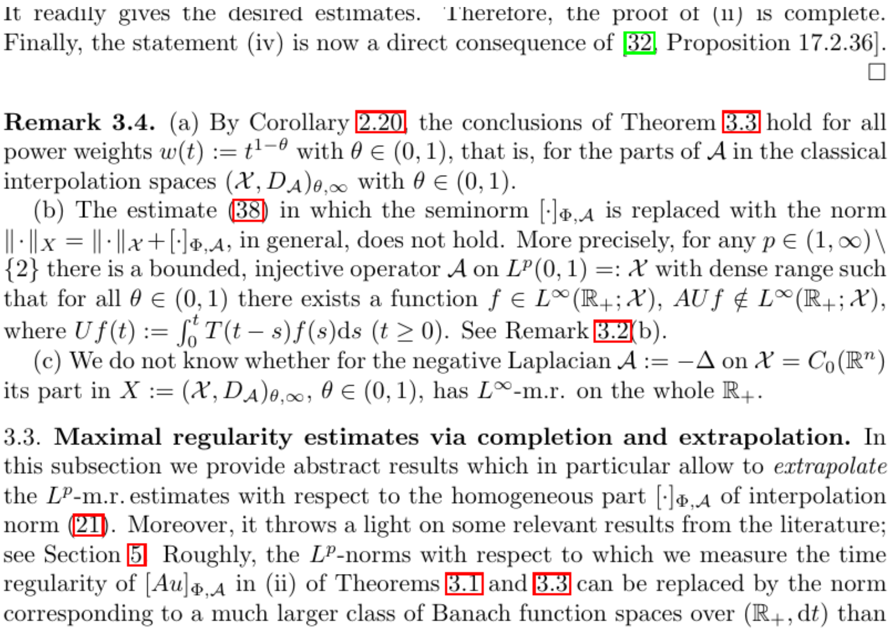
the part A of $\mathcal{A}$ in X_θ is given by

$$\mathrm{D}(A) = \{f \in \mathrm{D}(\mathcal{A}) \cap X_\theta : \mathcal{A}f \in X_\theta\}, \qquad Af = \mathcal{A}f.$$

2 Proof intuition

A single dilated bump cannot disprove maximal regularity: both sides of the relevant semigroup characterization scale in the same way. The obstruction appears after stacking many *nested* low-frequency bumps. At the common center their values add, so the norm of the stack grows like the number of scales. At any fixed heat time, however, only the bumps whose spatial scale matches that time contribute substantially to tAH_t; the other scales form two convergent geometric tails. Thus $\sup_t \|tAH_tf\|$ remains uniformly bounded.

The interpolation space with exponent ∞ has a non-dense generator, so one cannot immediately invoke the standard Kalton–Portal characterization. This is resolved by passing to the closure of the domain. Global L^∞-maximal regularity would pass to this closed subspace, where the restricted heat semigroup is strongly continuous. Kalton–Portal then turns the multiscale construction into a contradiction.



It readily gives the desired estimates. Therefore, the proof of (ii) is complete. Finally, the statement (iv) is now a direct consequence of [32, Proposition 17.2.36]. □

Remark 3.4. (a) By Corollary 2.20, the conclusions of Theorem 3.3 hold for all power weights $w(t) := t^{1-\theta}$ with $\theta \in (0,1)$, that is, for the parts of $\mathcal{A}$ in the classical interpolation spaces $(\mathcal{X}, D_{\mathcal{A}})_{\theta,\infty}$ with $\theta \in (0,1)$.

(b) The estimate (38) in which the seminorm $[\cdot]_{\Phi,\mathcal{A}}$ is replaced with the norm $\|\cdot\|_X = \|\cdot\|_{\mathcal{X}} + [\cdot]_{\Phi,\mathcal{A}}$, in general, does not hold. More precisely, for any $p \in (1,\infty) \setminus \{2\}$ there is a bounded, injective operator $\mathcal{A}$ on $L^p(0,1) =: \mathcal{X}$ with dense range such that for all $\theta \in (0,1)$ there exists a function $f \in L^\infty(\mathbb{R}_+;\mathcal{X})$, $AUf \notin L^\infty(\mathbb{R}_+;\mathcal{X})$, where $Uf(t) := \int_0^t T(t-s)f(s)\mathrm{d}s$ $(t \geq 0)$. See Remark 3.2(b).

(c) We do not know whether for the negative Laplacian $\mathcal{A} := -\Delta$ on $\mathcal{X} = C_0(\mathbb{R}^n)$ its part in $X := (\mathcal{X}, D_{\mathcal{A}})_{\theta,\infty}$, $\theta \in (0,1)$, has L^∞-m.r. on the whole $\mathbb{R}_+$.

3.3. **Maximal regularity estimates via completion and extrapolation.** In this subsection we provide abstract results which in particular allow to *extrapolate* the L^p-m.r. estimates with respect to the homogeneous part $[\cdot]_{\Phi,\mathcal{A}}$ of interpolation norm (21). Moreover, it throws a light on some relevant results from the literature; see Section 5. Roughly, the L^p-norms with respect to which we measure the time regularity of $[Au]_{\Phi,\mathcal{A}}$ in (ii) of Theorems 3.1 and 3.3 can be replaced by the norm corresponding to a much larger class of Banach function spaces over $(\mathbb{R}_+, \mathrm{d}t)$ than

Figure 1: Remark 3.4(c), at the foot of page 40 of arXiv:2502.16521v1.

3 Preliminaries

The semigroup representation of the real interpolation norm used in the source gives the equivalent norm

$$\|f\|_{\theta,*} := \|f\|_\infty + \sup_{s>0} s^{1-\theta} \|\mathcal{A}H_s f\|_\infty \qquad (f \in X_\theta). \tag{1}$$

Maximal regularity and the Kalton–Portal inequality are unchanged by an equivalent renorming, so we use (1) throughout.

We first isolate the non-dense-domain step.

Lemma 1 (passage to the closure of the domain)**.** *Let A be a closed operator on a Banach space X, and put $Y = \overline{\mathrm{D}(A)}^X$. Let A_0 be the part of A in Y. Suppose that $-A_0$ generates the restriction to Y of the semigroup generated by $-A$. If A has L^∞-maximal regularity on $\mathbb{R}_+$, then so does A_0.*

Proof. Take $f \in L^\infty(\mathbb{R}_+;Y)$ and let u be the strong solution in X of $u' + Au = f$, $u(0) = 0$. Since $u(t) \in \mathrm{D}(A) \subset Y$ for almost every t and u is continuous as an X-valued function, closedness of Y gives $u(t) \in Y$ for every t. The derivative of an absolutely continuous X-valued function whose values lie in the closed subspace Y belongs to Y almost everywhere: this follows directly from the difference quotients. Hence $u'(t) \in Y$ and

$$Au(t) = f(t) - u'(t) \in Y$$

almost everywhere. Thus $u(t) \in \mathrm{D}(A_0)$ almost everywhere and it is the strong solution for A_0. The L^∞ estimate is the same because Y has the inherited norm, and uniqueness in Y follows from uniqueness in X. □

We will also use a standard geometric-sum observation.

Lemma 2 (dyadic profiles)**.** *If $a, b > 0$ and $q > 1$, then*

$$\sup_{v>0} \sum_{j=1}^{\infty} \min\{(v/q^j)^a, (v/q^j)^{-b}\} < \infty.$$

Proof. Split the indices at the last j for which $v/q^j \geq 1$. On either side the summands are bounded by a geometric progression, with ratios q^{-b} and q^{-a}, respectively. The same proof covers the cases in which one of the two index sets is empty. $\square$

4 The multiscale obstruction

Choose $0 \leq \varphi \in C_c^\infty(\mathbb{R}^n)$ with $\varphi(0) = 1$. For $m = 0, 1, 2$ put

$$F_m(u) = \|\mathcal{A}^m H_u \varphi\|_\infty, \qquad u > 0.$$

There are constants C_m such that

$$F_m(u) \leq C_m \min\{1, u^{-m-n/2}\}, \qquad u > 0. \tag{2}$$

Indeed, for $0 < u \leq 1$, commutation with $\mathcal{A}^m$ and heat contractivity give $F_m(u) \leq \|\mathcal{A}^m \varphi\|_\infty$. For $u \geq 1$, convolution with the m-th Laplacian of the Gaussian heat kernel gives

$$F_m(u) \leq \|\mathcal{A}^m k_u\|_\infty \|\varphi\|_1 \leq C_m u^{-m-n/2}.$$

Set

$$R_j = 2^j, \qquad r_j = R_j^2 = 4^j, \qquad \varphi_j(x) = \varphi(x/R_j), \qquad x_N = \sum_{j=1}^N \varphi_j.$$

Every φ_j belongs to $C_c^\infty(\mathbb{R}^n)$, and both it and all its Laplacians belong to X_θ. Hence $x_N \in \mathrm{D}(A)$. On the other hand,

$$\|x_N\|_{\theta,*} \geq \|x_N\|_\infty \geq x_N(0) = N. \tag{3}$$

Heat scaling gives, for $m = 0, 1, 2$,

$$\|\mathcal{A}^m H_t \varphi_j\|_\infty = r_j^{-m} F_m(t/r_j). \tag{4}$$

The base part of the thermic norm of $tAH_t x_N$ is therefore bounded by

$$\begin{aligned} \|tAH_t x_N\|_\infty &\leq \sum_{j=1}^N (t/r_j) F_1(t/r_j) \\ &\leq C \sum_{j=1}^N \min\{t/r_j, (t/r_j)^{-n/2}\} \leq C, \end{aligned} \tag{5}$$

uniformly in $t > 0$ and N, by Lemma 2.

For the homogeneous part, fix $s, t > 0$. Semigroup commutation and (4) yield

$$s^{1-\theta} \|\mathcal{A} H_s [tAH_t \varphi_j]\|_\infty = s^{1-\theta} t r_j^{-2} F_2((s+t)/r_j).$$

Put $u = (s+t)/r_j$. Since $s^{1-\theta} t \leq (s+t)^{2-\theta}$ and $r_j \geq 1$, the last expression is at most

$$r_j^{-\theta} u^{2-\theta} F_2(u) \leq C \min\{u^{2-\theta}, u^{-n/2-\theta}\}.$$

Both exponents in Lemma 2 are positive. Summing over j and then taking the supremum over s proves, together with (5), that

$$\sup_{t>0} \|tAH_t x_N\|_{\theta,*} \leq C_{n,\theta,\varphi} \qquad \text{for every } N. \tag{6}$$

The large-time term in the Kalton–Portal condition vanishes. For each fixed j,

$$\|H_t \varphi_j\|_\infty \leq C R_j^n t^{-n/2} \longrightarrow 0.$$

Moreover, once $t \geq r_j$, (2) gives

$$\sup_{s>0} s^{1-\theta}\|\mathcal{A}H_{s+t}\varphi_j\|_\infty \leq C \sup_{s>0} s^{1-\theta} r_j^{-1}((s+t)/r_j)^{-1-n/2} \leq C_{n,\theta} R_j^n t^{-\theta-n/2} \longrightarrow 0.$$

Since N is fixed, summing proves

$$\lim_{t\to\infty} \|H_t x_N\|_{\theta,*} = 0. \tag{7}$$

5 Full negative answer

Theorem 3. *For every $n \geq 1$ and every $\theta \in (0,1)$, the part A of $\mathcal{A} = -\Delta$ in*

$$X_\theta = (C_0(\mathbb{R}^n), \mathrm{D}(\mathcal{A}))_{\theta,\infty}$$

does not have L^∞-maximal regularity on $\mathbb{R}_+$.

Proof. Suppose to the contrary that A has global L^∞-maximal regularity. Let

$$Y = \overline{\mathrm{D}(A)}^{X_\theta}$$

with the inherited thermic norm, and let A_0 be the part of A in Y. The source proves that A generates the bounded analytic heat semigroup on X_θ and that its restriction to Y is a bounded analytic C_0-semigroup with generator $-A_0$. By Lemma 1, A_0 has L^∞-maximal regularity on $\mathbb{R}_+$.

The Kalton–Portal characterization [2, Theorem 3.7], also recorded as Lemma 2.13 of the source, now supplies a constant C such that every $y \in Y$ satisfies

$$\|y\|_{\theta,*} \leq C \sup_{t>0} \|tA_0H_ty\|_{\theta,*} + \limsup_{t\to\infty} \|H_ty\|_{\theta,*}. \tag{8}$$

Each x_N lies in $\mathrm{D}(A) \subset Y$. Applying (8) to x_N and using (3), (6), and (7) gives

$$N \leq CC_{n,\theta,\varphi}$$

for every N, a contradiction. □

Corollary 4 (answer to Remark 3.4(c))**.** *The answer to the question in arXiv:2502.16521 is no for every dimension and every interpolation parameter covered by the question.*

6 Scope, novelty, and verification

The proof concerns the *big* real interpolation space with fine index ∞ and global time $\mathbb{R}_+$. It does not contradict the finite-interval Da Prato–Grisvard theorem, nor the source's homogeneous-seminorm estimates. The obstruction is specifically the accumulation of arbitrarily many low frequencies in the inhomogeneous C_0 component of the norm.

The run's cheap indexes and parsed corpus were searched for the arXiv id, the exact question, the interpolation space over $C_0(\mathbb{R}^n)$, and global L^∞ maximal regularity. Exact-title and exact-question searches through 11 August 2026 recovered the source but no later primary-source resolution. Because an unindexed or differently phrased antecedent remains possible, novelty confidence is moderate; the mathematical claim is submitted as a candidate full solution.

The argument is analytic and uses no computational evidence. Review should focus on Lemma 1, the powers of r_j in (4), and the use of the thermic norm in the large-time estimate. All three points have been expanded explicitly above.

References

[1] S. Król, M. Mastyło, and J. Sarnowski, *Maximal regularity estimates for the abstract Cauchy problems*, arXiv:2502.16521v1, 2025.

[2] N. J. Kalton and P. Portal, *Remarks on ℓ_1 and ℓ_∞-maximal regularity for power-bounded operators*, J. Aust. Math. Soc. **84** (2008), 345–365.